

CONTROLLED VGGT STUDY

From Real-Scene Robustness to Synthetic Ground-Truth Adaptation

Version 3 seminar report | VGGT, CVPR 2025

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Abstract

We reproduce the maintained official VGGT-1B implementation and study its behavior on two calibrated ETH3D scenes and a held-out synthetic TartanAir trajectory. The real-scene study varies overlapping input views from 2 to 10. The synthetic extension adds exact camera poses and metric depth, enabling similarity-aligned camera and scale-aligned depth evaluation. On TartanAir P000, the official checkpoint achieves 3.14% depth AbsRel at two views and 6.89% at ten views, while mean camera rotation error remains below 0.77 degrees. A leakage-safe, 30-step frozen-aggregator adaptation is operationally feasible on an RTX 5080 but worsens held-out AbsRel and delta-1, demonstrating over-specialization rather than general improvement. This negative result is retained as evidence about fine-tuning risk.

Assignment brief

את התרגיל הזה יש לבסס על המאמר שהצגתם.
 אם אתם זוג תמצאו לזה פתרון (משהו משותף לשתי העבודות או התרכזות באחת מהם).
 יש לנסות את המערכת על קלטים שונים ולחפש חוזקות וחולשות.
 אם ניתן לעשות למערכת finetuning זה גם טוב.
 מה שיש להגיש זה מסמך או מצגת שמתאר מה עשיתם.
 בהצלחה, אילן

Requirement	Evidence in this submission
Based on the presented paper	Official VGGT code/checkpoint reproduction and paper-grounded analysis
Shared direction for a pair	Joint focus on robustness, synthetic accuracy, and adaptation
Different inputs	ETH3D indoor/outdoor plus synthetic TartanAir, 2/4/6/8/10 views
Strengths and weaknesses	Measured geometry, confidence, memory, failure and overfitting analysis
Fine-tuning if feasible	Measured RTX 5080 probe, 30-step head adaptation, held-out comparison
Document or presentation	This illustrated report and reproducible project artifacts

1. VGGT and the study design

VGGT predicts camera parameters, depth, point maps, confidence, and tracks from one or more same-scene RGB images in a single feed-forward model. DINOv2 image tokens enter an aggregator alternating frame-local and global attention; task-specific heads decode geometry.

VGGT IN BRIEF

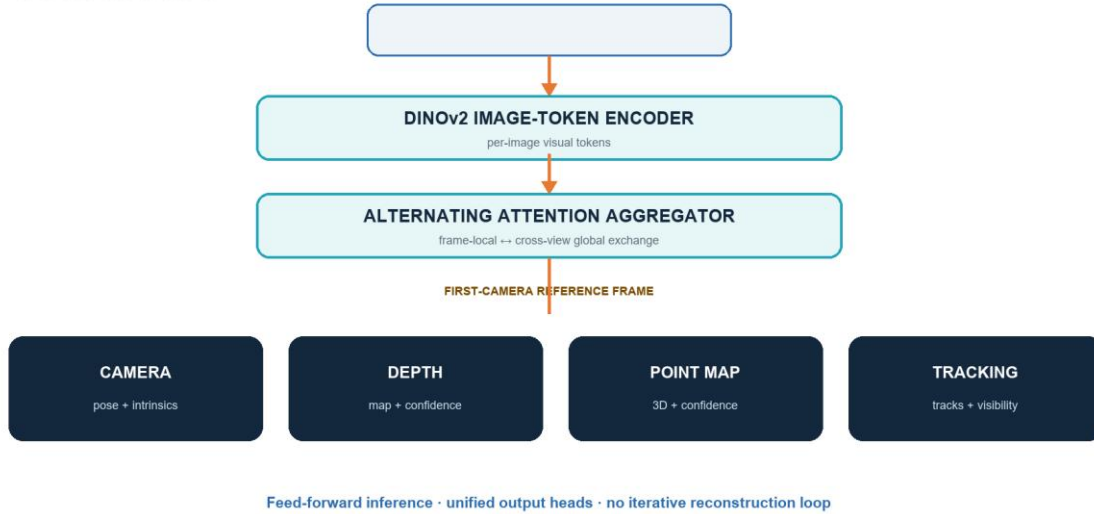


Figure 1. Project-authored summary of the VGGT architecture.

The experimental ladder separates operational reproduction, controlled real-scene view-count testing, held-out synthetic scoring, and bounded adaptation. Predictions are never compared in raw scale: camera centers use Sim(3) alignment and depth uses a global median scale.

2. Inputs and frozen protocol

The existing study uses ETH3D delivery_area and courtyard with deterministic overlap-aware nested subsets. V3 adds TartanAir V2 ArchVizTinyHouseDay/easy. TartanAir is not listed in the VGGT training inventory recorded from the paper, although unreported overlap cannot be excluded.

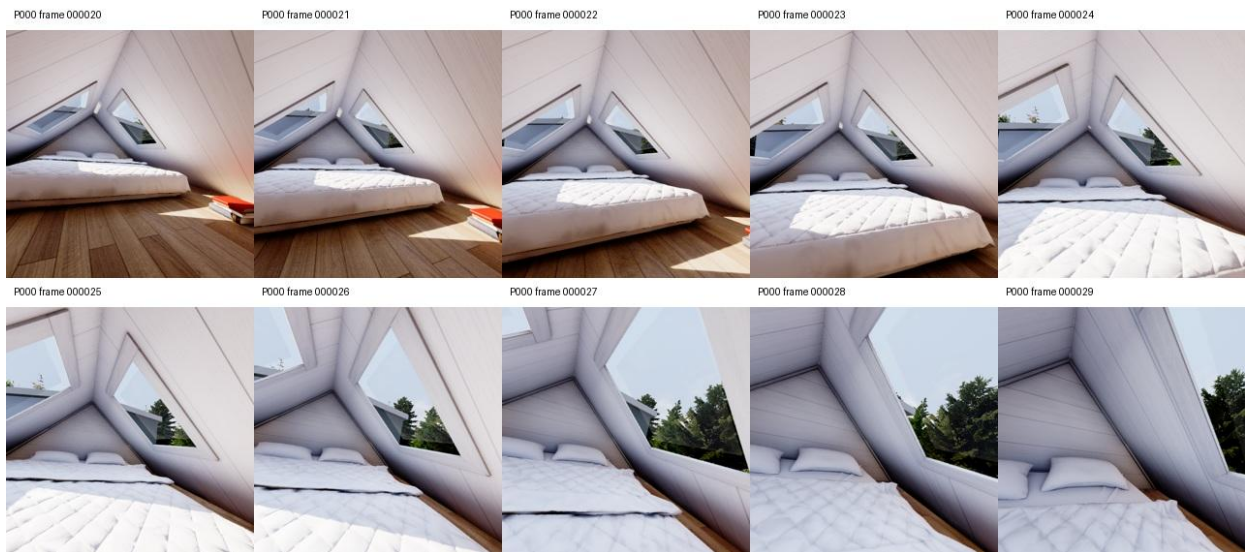


Figure 2. Frozen P000 frames 20-29. Adjacent pairs retain at least 136 RANSAC inliers across 1.91 m of ground-truth path motion.

Role	Trajectories	Use
Training	P001-P005	Only for the bounded adaptation
Validation	P006	Checkpoint selection; best at step 15
Test	P000	Untouched baseline/adapted comparison

3. Pretrained synthetic baseline

Views	AbsRel	delta-1	Rot. deg	Conf/error rho	Forward	GiB
S2	3.14%	0.978	0.089	-0.641	0.545s	5.15
S4	3.55%	0.972	0.361	-0.467	0.351s	5.68
S6	3.73%	0.966	0.601	-0.546	0.548s	6.35
S8	5.29%	0.956	0.700	-0.465	0.785s	7.02
S10	6.89%	0.949	0.767	+0.066	1.073s	7.11

The checkpoint is highly accurate for this short indoor window, but adding views does not monotonically help. Depth AbsRel more than doubles from S2 to S10, while camera rotation remains below one degree. Confidence is useful when its Spearman correlation with error is negative; at S10 the sign becomes slightly positive, exposing a calibration weakness.

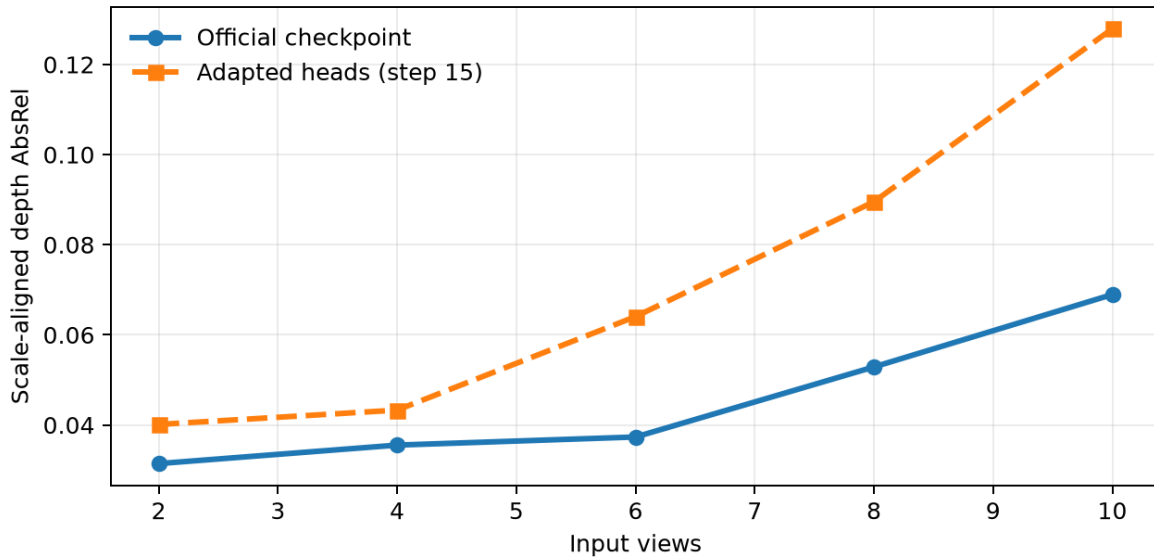


Figure 3. Held-out scale-aligned depth error before and after adaptation.

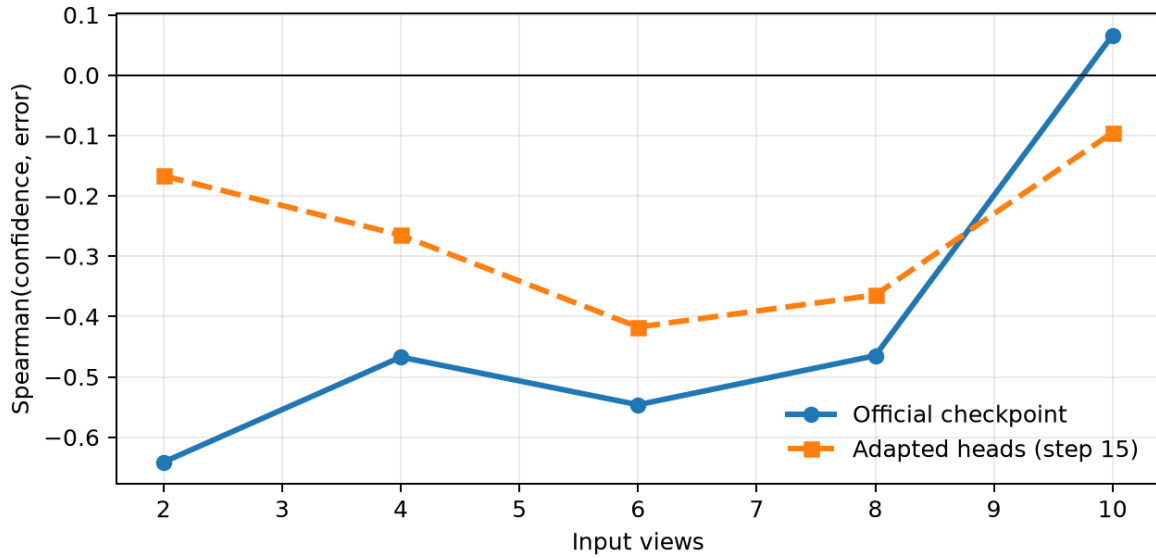


Figure 4. Confidence calibration weakens as more views are added.

4. Fine-tuning feasibility and execution

The maintained repository's training code is a post-publication reimplementaion. Our bounded experiment freezes the 1.2B aggregator and updates only camera and depth heads using adjacent two-view samples. It uses BF16, AdamW, gradient clipping, 30 optimizer steps, and P006-only checkpoint selection. This is domain adaptation, not reproduction of the paper's 64-A100 training.

Feasibility item	Measured result
GPU	NVIDIA GeForce RTX 5080, 15.92 GiB
Forward/backward probe	5.95 GiB allocated; 6.52 GiB reserved
Optimizer steps	30; camera and depth heads only
Elapsed adaptation time	25.8 seconds
Checkpoint selection	P006 validation minimum at step 15
Test isolation	P000 never used for training or selection

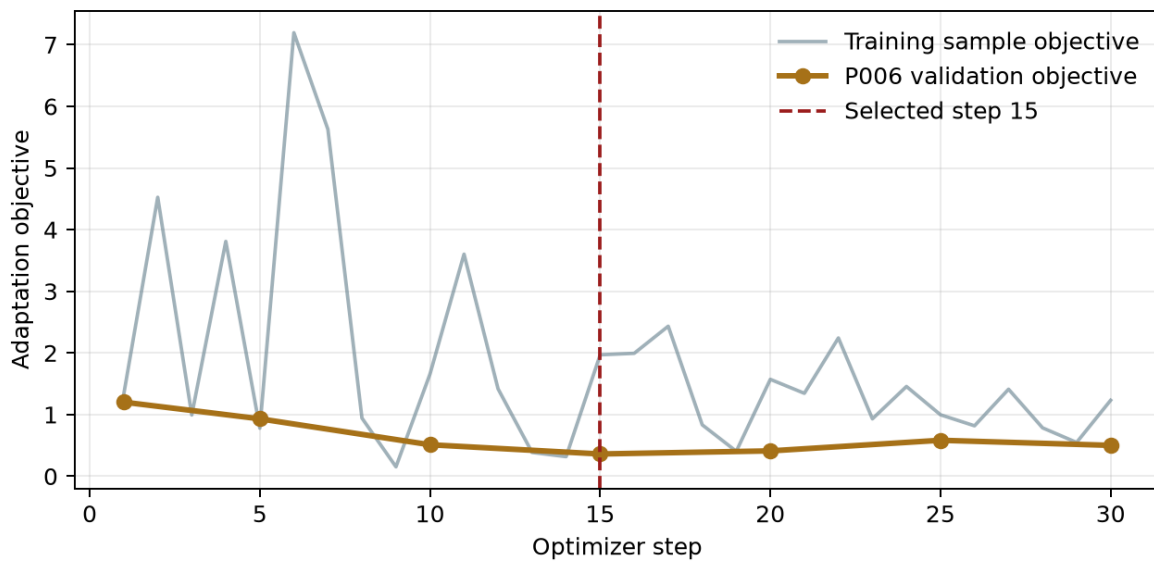


Figure 5. Validation improves until step 15, then begins to regress.

5. Pretrained versus adapted

Views	Base AbsRel	Adapt AbsRel	Base RMSE	Adapt RMSE	Base rot.	Adapt rot.
S2	3.14%	4.01%	2.183	2.107	0.089	0.122
S4	3.55%	4.33%	2.635	2.565	0.361	0.374
S6	3.73%	6.40%	3.790	3.677	0.601	0.619
S8	5.29%	8.95%	4.458	4.436	0.700	0.712
S10	6.89%	12.79%	4.259	4.225	0.767	0.773

Adaptation is not a held-out success. RMSE falls slightly at every view count, but AbsRel and delta-1 worsen, and camera errors rise modestly. At S10, AbsRel increases from 6.89% to 12.79%. The most defensible interpretation is that a tiny two-view adaptation changes the error distribution and over-specializes to the training/validation trajectories. The official checkpoint remains the recommended model.

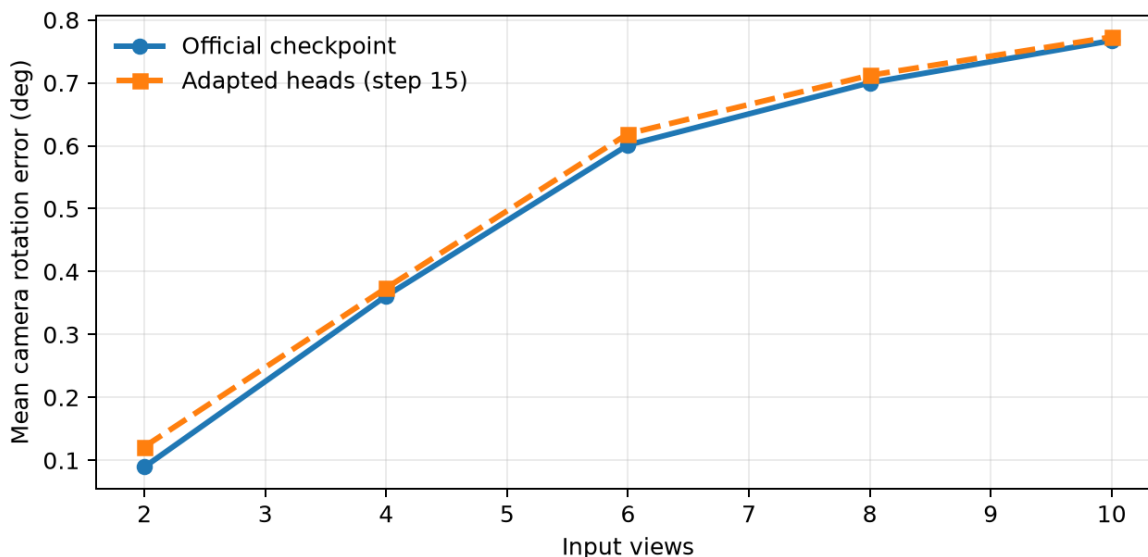


Figure 6. Camera rotation error remains low but does not improve after adaptation.

6. Strengths, weaknesses, and lessons

Observed strengths

- All official output heads operated reliably in the earlier reproduction and real-scene pilots.
- On synthetic P000, camera orientation stays below 0.77 degrees mean error through ten views.
- Two-view scale-aligned depth is strong: 3.14% AbsRel and 97.8% delta-1.
- Frozen-head adaptation is practical on a consumer RTX 5080 with substantial memory margin.

Observed weaknesses

- More overlapping views do not guarantee lower depth error; S10 is worse than S2.
- Confidence calibration becomes unreliable at S10 even when outputs remain visually plausible.
- Two-point Sim(3) alignment makes S2 camera ATE degenerate; rotation and larger subsets are more informative.
- Short domain adaptation can improve one metric while harming more interpretable relative metrics.

Methodological lesson

A fine-tuning run should not be labeled successful because its training loss decreases. A held-out split, frozen baseline, multiple metrics, and retained negative results are essential.

7. Limitations

The quantitative synthetic study covers one environment and one ten-frame test window. TartanAir is absent from the documented VGGT training mix, but unreported overlap cannot be excluded. Depth validity is limited to 0.1-100 m to remove synthetic far-value sentinels. The fine-tune uses only 30 steps

and adjacent two-view samples; it does not establish the best possible adaptation recipe. Timings are single synchronized forwards, not repeated benchmarks. The original ETH3D study remains qualitative because laser alignment was outside its frozen scope.

8. Conclusion

VGGT is operationally strong and geometrically accurate on the selected synthetic sequence, but view-count gains are not monotonic and confidence can cease to track error. Fine-tuning the camera/depth heads is feasible on an RTX 5080, yet the measured adaptation overfits and should not replace the official checkpoint. The project therefore satisfies the assignment through varied inputs, explicit strengths and weaknesses, and an honest measured fine-tuning study.

References

1. Wang et al. VGGT: Visual Geometry Grounded Transformer. CVPR, 2025.
2. Schops et al. A Multi-View Stereo Benchmark with High-Resolution Images and Multi-Camera Videos. CVPR, 2017.
3. Wang et al. TartanAir: A Dataset to Push the Limits of Visual SLAM. IROS, 2020.
4. Official VGGT repository, pinned project revision a288dd0f14786c93483e45524328726ab7b1b4ce.
5. TartanAir V2 documentation and official theairlabcmu/tartanair2 distribution.